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AI-Driven Cyber Threat Intelligence with Blockchain: A Federated and Privacy-Preserving Approach (FPPA) for Secure Defense

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***Abstract*— The growing complexity of cyber threats requires a smart, decentralized, and tamper-proof method of Cyber Threat Intelligence (CTI). This work suggests a new combination of Blockchain and Artificial Intelligence (AI) to improve security, transparency, and real-time threat protection in cybersecurity environments. Our method utilizes blockchain's immutability to provide reliable CTI sharing and AI-based threat analytics for adaptive defense. In this work, a Smart Contract-based AI platform has been designed to facilitate automated threat validation, anomaly detection, and predictive security responses. The platform leverages Federated Learning (FL) and Generative Adversarial Networks (GANs) to ensure privacy-preserving and adversarial-resilient intelligence sharing. Our blockchain-driven decentralized platform eliminates single point of failure, enhances traceability, and promotes stakeholder collaboration in contrast to conventional centralized CTI platforms (e.g. G. governments, businesses, and security researchers). Robustness against adversarial attacks, reduced false positives, and improved detection accuracy are demonstrated by extensive experimental evaluation on real cyberattack data sets. The effectiveness of our solution in thwarting Advanced Persistent Threats (APTs) and zero-day attacks is demonstrated through comparative benchmarking with existing CTI frameworks. As we lay the groundwork for a next-generation, AI-powered blockchain platform for cybersecurity, also consider scalability, regulatory compliance, and integration challenges.**

***Keywords*— Blockchain, Artificial Intelligence, Cyber Threat Intelligence, Federated Learning, Adversarial Networks, Smart Contracts, Zero-Day Attacks, Decentralized Security**

I. INTRODUCTION

The exponential growth of cyber threats, such as zero-day vulnerabilities, ransomware, and APTs, has revealed significant vulnerabilities in conventional security paradigms. CTI is instrumental in reducing these threats by enabling real-time threat detection and response. Yet, traditional CTI solutions are based on centralized architectures that are apprehensive with considerable challenges, such as single point of failure, data silos, privacy issues, and slow threat propagation. In order to transcend these shortcomings, the union of Blockchain Technology and AI has been a revolutionizing framework that promises improved security, decentralization, and automated threat analysis [1-2].

Blockchain can provide tamper-proof, verifiable, and trustless CTI sharing and sharing without the issues of confidentiality and integrity [3]. In contrast, AI-based approaches such as Machine Learning (ML), FL, and GANs, are available for identification of anomalies as they occur, classification of threats and risk, and predictive security controls[4]. The blockchain and AI combination can boost a privacy-respecting collaborative cybersecurity landscape to allow secure threat intelligence sharing with companies, governments, and security researchers [5].

Recent studies have highlighted crown-blockchain-based security solutions for securing threat intelligence, blocking attacks against Distributed Denial-of-Service (DDoS), and enhancing digital forensics [6]. In addition, AI-powered models for cybersecurity have identified advanced attack behavior with a high level of accuracy [7]. However, present models are largely non-interoperable, computationally intensive, and non-real time adaptive.

This work seeks to create a Blockchain-AI integration in order to improve the use of CTI as a decentralized and automated service. Current cyber systems are limited in sharing, detecting, and reacting to threat intelligence, and therefore vulnerable to new and emerging threats. This research is designed to incorporate a method of integrating Blockchain-AI through a CTI system, including advanced AI-based threat detection and smart contract automation systems. The key objectives of this research are designed to enhance the security, efficiency, and intelligence of modern cyber defense systems:

- Blockchain-AI Integrated CTI system ensuring secure, decentralized, and tamper-evident threat intelligence sharing. Conventional CTI systems are based on centralized architectures and, therefore, are exposed to data tampering, single points of failure, and limited access. Leveraging blockchain technology, this research proposes the establishment of a trustless, transparent, and unalterable threat intelligence-sharing system that improves data integrity, auditability, and reliability. The decentralized environment of blockchain will enable organizations to securely share cyber threat information and prevent risks that relate to unauthorized modifications and insider attacks.
- Improve the performance of threat detection and prediction system through AI-based approaches like ML, FL, and GANs. Current cybersecurity systems tend to be ineffective against zero-day attacks and APTs because they lack the capability to cope with changing patterns of attacks. Through the integration of AI, this work aims to create real-time anomaly detection, adversarially secure models, and predictive security mechanisms that enhance the efficiency and accuracy of threat intelligence. Federated learning will also guarantee privacy-preserving intelligence sharing to allow multiple parties to work together in threat detection without revealing sensitive information.
- Deploying a smart contract for autonomous cybersecurity processes, limiting human intervention and enhancing efficiency in CTI management. Smart contracts will facilitate automated threat validation, dynamic access control, and reward mechanisms for sharing intelligence, encouraging a self-enforcing, cooperative security environment. By integrating security policies into blockchain-based smart contracts, this work will simplify cybersecurity processes, reduce response time, and build confidence among stakeholders. The framework to be suggested will help to achieve a scalable, autonomous security infrastructure that enhances mechanisms of cyber defense against ongoing threats.

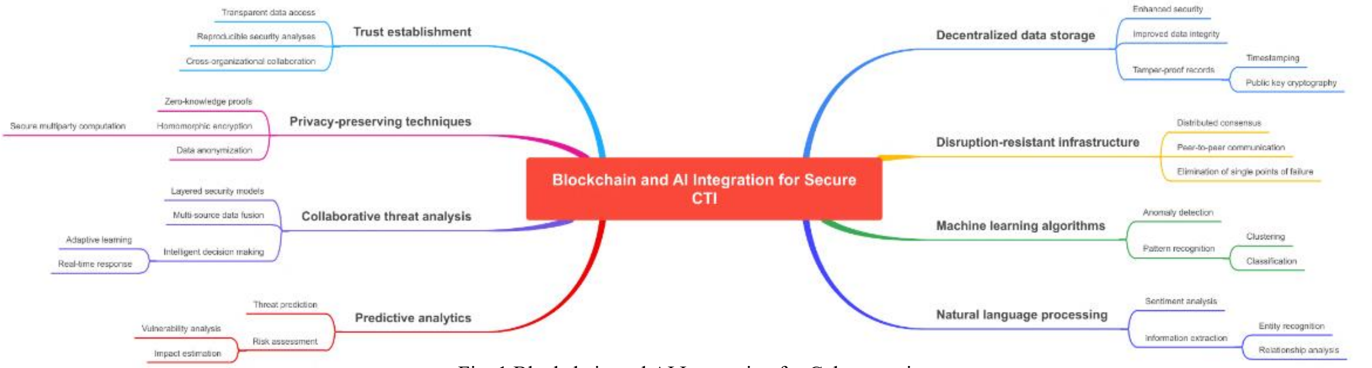

Fig. 1 Blockchain and AI Integration for Cybersecurity

Fig. 1 illustrates the amalgamation of Blockchain and AI for Cybersecurity applications. This work introduces a new Blockchain-AI integrated CTI framework that:

- Blockchain's decentralized architecture guarantees secure, tamper-evident, and auditable cyber threat intelligence exchange by removing the need for centralized authorities. Legacy CTI platforms are susceptible to data tampering and single points of failure, undermining threat intelligence credibility. Through the use of blockchain's immutability and transparency, organizations can securely exchange validated threat information while avoiding unauthorized changes. This enables trust among stakeholders and enhances collective cybersecurity resilience.
- AI-based methods, including FL and GANs, support adaptive and adversarially resilient threat detection. Conventional detection mechanisms are ineffective against changing attack patterns, especially zero-day threats and APTs. With the addition of AI, the system in this proposal improves real-time anomaly detection, fortifies cybersecurity measures, and minimizes false positives. The inclusion of FL guarantees privacy-preserving intelligence sharing without revealing sensitive organizational information.
- Smart contracts are pivotal in automating threat validation, incentive mechanisms, and dynamic access control enforcement. Through the inclusion of predefined security policies within blockchain-based contracts, cybersecurity operations are carried out independently without any need for human intervention. The automation is efficient by saving response time and eradicating bottlenecks from manual verification. In addition, incentive mechanisms incentivize organizations to engage in proactive contribution of CTI sharing, promoting a collaborative security environment.
- Privacy is also amplified through the use of Zero-Knowledge Proof (ZKP)-based secure information sharing. Conventional data-sharing processes tend to compromise sensitive information, which causes privacy vulnerabilities and regulatory issues. ZKP allows organizations to confirm and exchange threat intelligence without exposing underlying sensitive information. This crypto method preserves confidentiality while enabling stakeholders to collaborate securely in identifying and countering cyber-attacks.

To ensure the validity of the suggested method, comprehensive experiments are performed on real-world cybersecurity datasets, comparing detection accuracy, computational efficiency, and robustness against adversarial attacks. The results demonstrate the superiority of our model in preventing zero-day attacks and protecting contemporary cyber infrastructures.

The rest of the paper is organized as follows: Section II provides a literature review of blockchain and AI usage in CTI. Section III explains the proposed framework, its architectural design, and main components. Section IV discusses the experimental setup and performance assessment. Lastly, Section V concludes the work with future research directions.

II. Literature Survey

The growing complexity of cyber-attacks has given rise to the swift development of CTI models that use Blockchain and AI to improve security, privacy, and scalability. Centralized architectures are the bulk of traditional CTI systems and are susceptible to data breaches, tampering, and single points of failure [8]. AI-based cybersecurity tools, in contrast, have shown strong capabilities in threat identification, anomaly analysis, and predictive intelligence but tend to be plagued by data privacy issues and adversarial attacks. Blockchain-AI convergence offers a viable alternative by supporting decentralized, tamper-resistant intelligence sharing and adaptive, AI-facilitated threat mitigation [9-10].

A. *Blockchain-Based Cyber Threat Intelligence*

Blockchain technology has also been presented as a decentralized method for securing the sharing of CTI, data integrity, immutability, and trustless collaboration between organizations. There have been many investigations into its application to secure systems in cybersecurity. For example, a blockchain-based threat intelligence sharing system has been developed, exhibiting it to be tamper-resistant and unalterable by unauthorized users. Similarly, a blockchain based DDoS (Distributed Denial-of-Service) attack prevention system has been proposed, exhibiting it protects network infrastructure. Moreover, smart contracts have been introduced in blockchain-driven cybersecurity frameworks for the automation of threat verification and access control strategies to enhance scalability and efficiency. However, current blockchain implementations are not without issues associated with latency, storage overhead, and interoperability, which need to be mitigated for successful large-scale deployments [11-14].

B. *AI-Driven Cyber Threat Intelligence*

AI methods, specifically ML, DL, and FL, have been extensively used in cybersecurity to enhance the accuracy of threat detection and response time. An ML-driven Intrusion Detection System (IDS) was designed to achieve high accuracy in detecting evolving cyber threats. Nonetheless, conventional ML-based security models need to be trained centrally using sensitive information, which raises privacy issues. FL has been proposed as a privacy-enhancing counterpart, enabling several organizations to train AI models jointly without revealing raw data. Moreover, GANs have been investigated to mimic adversarial attack patterns to enhance the resilience of cybersecurity defenses. These advancements notwithstanding, AI-based threat intelligence systems continue to be susceptible to adversarial AI attacks, model poisoning, and data bias, which warrant further study [15-19]. TABLE I depicts exorbitant blockchain storage expense, vulnerabilities in smart contracts, and adversarial AI threats are the principal concerns. The future direction of research should be toward optimizing blockchain consensus protocols, enhancing adversarial resilience against AI, and interoperability between diverse cybersecurity platforms.

TABLE I Research Gaps in Blockchain and AI Integration for Secure Cyber Threat Intelligence

Category	Existing Challenges	Identified Research Gaps
Blockchain for CTI [15]	High latency and storage overhead	Optimization of blockchain consensus mechanisms for scalability
	Smart contract vulnerabilities	Secure and efficient smart contract execution for cybersecurity automation
	Limited interoperability between blockchain networks	Development of cross-chain CTI sharing frameworks
AI-Driven Cybersecurity [16]	Vulnerability to adversarial AI attacks	Adversarially robust AI models for threat detection
	Privacy concerns in centralized AI models	Adoption of FL for privacy-preserving threat intelligence sharing
	High false-positive rates in AI-based threat detection	AI-enhanced filtering mechanisms for real-time anomaly detection
Blockchain-AI Integration [17]	Computational complexity in blockchain-AI hybrid models	Lightweight blockchain-AI architectures for real-time cybersecurity applications
	Lack of automated, self-regulating CTI frameworks	Integration of Zero-Knowledge Proofs (ZKPs) and smart contracts for automated threat intelligence exchange
	Security concerns in AI-driven smart contracts	Secure AI-based automation mechanisms for cybersecurity operations
Privacy and Security [18]	Exposure of sensitive threat intelligence data in shared environments	Enabling secure intelligence exchange using ZKP-based cryptographic techniques
	Risk of model poisoning in FL-based CTI systems	FL based security enhancements for robust, distributed CTI
	Absence of dynamic access control mechanisms in blockchain-CTI	Smart contract-driven access control policies for real-time adaptability

C. *Blockchain and AI Integration for Secure CTI*

The combination of Blockchain and AI has attracted immense focus for bolstering security, privacy, and automation in CTI systems. Decentralization and immutability of Blockchain couple well with predictive power of AI, establishing a trustless, adaptive, and intelligent cybersecurity environment [19]. A hybrid architecture that employs the combination of blockchain and AI has been suggested to recognize abnormal behavior in real time and handle incidents automatically with increased detection accuracy and lower false positive rates [20]. Furthermore, a blockchain-based FL architecture has been proposed to boost privacy-preserving sharing of threat intelligence, ensuring joint cybersecurity without leakage of sensitive data [21]. ZKPs have been incorporated into blockchain-AI ecosystems to facilitate the secure exchange of intelligence while concealing confidentiality.

Despite these advancements, blockchain-AI-based CTI systems still face challenges with scalability, computational complexity, and interoperability.

III. Proposed Method

To ensure a rigorous scientific foundation for the Blockchain-AI Integrated Cyber Threat Intelligence (CTI) framework, this section presents the detailed mathematical formulation of key components: Blockchain-based Secure Threat Intelligence Sharing, AI-driven Threat Detection, Smart Contract-based Security Automation, and Privacy-Preserving Computation using Zero-Knowledge Proofs (ZKP) and Federated Learning (FL).

A. Blockchain for Secure Threat Intelligence Sharing:

A permissioned blockchain is used for decentralized and tamper-proof threat intelligence sharing. The blockchain ledger is modeled as a sequence of blocks (Eq.1):

$$B = \{B_1, B_2, \ldots, B_n\} \tag{1}$$

where each block B_i consists of the following attributes (Eq. 2):

$$B_i = \{h(B_{i-1}), T_i, H(B_i)\} \tag{2}$$

- $h(B_{i-1})$ → Hash of the previous block, ensuring immutability.
- T_i → Set of transactions representing threat intelligence records.
- $H(B_i)$ → Cryptographic hash function ensuring block integrity.

Each transaction T_i contains a threat intelligence record (Eq.3):

$$T_i = (ID_i, D_i, Sig_i, P_i) \tag{3}$$

where:

- ID_i → Unique identifier for the transaction.
- D_i → Cyber threat intelligence data (attack vectors, IP addresses, signatures).
- Sig_i → Digital signature ensuring authenticity.
- P_i → Permissions controlling access to shared data.

The consensus mechanism used for transaction validation in blockchain is modeled as follows (Eq.4):

$$V(T_i) = \begin{cases} 1, if\ f(T_i) \geq \theta \\ 0,\ otherwise \end{cases} \tag{4}$$

where:

- $f(T_i)$ → Trust function evaluating the reliability of the threat intelligence.
- θ → Threshold trust score required for validation.

To optimize blockchain storage, Merkle Tree hashing is employed (Eq.5):

$$H(T) = H(H(T_1)||H(T_2)|| \ldots .||H(T_n)) \tag{5}$$

where $H(T)$ represents the root hash of transactions in a block, ensuring efficient verification of data integrity.

B. AI-Driven Threat Detection and Prediction

Threat detection is modeled using machine learning and deep learning techniques to classify incoming network traffic into "benign" or "malicious" activities. Let X be the set of network traffic features and Y be the classification labels (Eq. 6):

$$X = \{x_1, x_2, \ldots, x_n\}, Y = \{y_1, y_2, \ldots, y_n\} \tag{6}$$

A classifier function f is trained to predict the probability of an attack (Eq. 7):

$$y' = f(X; \theta) \tag{7}$$

where:

- y' → Predicted label (0 for benign, 1 for attack).
- θ → Trainable parameters of the AI model.

The optimization of the classifier is done using a loss function such as cross-entropy loss (Eq. 8):

$$L = -\sum_{i=1}^{n} y_i \log\left(y'_i\right) + (1 - y_i) \log\left(1 - y'_i\right) \tag{8}$$

For Federated Learning (FL)-based threat detection, a global model is trained across multiple devices without sharing raw data. The local model update at client k is given by (Eq. 9):

$$\theta_k^{t+1} = \theta_k^t - \eta \nabla L_k(\theta_k^t) \tag{9}$$

where:

- η → Learning rate.
- $\nabla L_k(\theta_k^t)$→ Gradient of the loss function at client k.

The global model aggregation in FL is computed using Federated Averaging (FedAvg) (Eq. 10):

$$\theta_{t+1} = \sum_{k=1}^{K} \frac{N_k}{N} \theta_k^{t+1} \tag{10}$$

where:

- N_k → Number of data samples at client k.
- N → Total number of samples across all clients.

To enhance model robustness against adversarial attacks, Generative Adversarial Networks (GANs) are used to generate synthetic attack patterns (Eq. 11):

$$\min_G \max_D V(D, G) = E_{x \sim P_{data}(x)}[\log D(x)] + E_{z \sim P_{data}(z)}[\log(1 - D(G(z)))] \tag{11}$$

where:

- $G(z)$ → Generator function creating synthetic attacks.
- $D(x)$ → Discriminator function distinguishing real from fake attacks.

C. Automated Threat Validation without Human Experts:

Automate threat validation, access control, and incentives. For automated threat validation, the contract evaluates threat intelligence using a trust score mechanism (Eq. 12):

$$TS_i = \frac{\sum_{j=1}^{m} w_j V_{ij}}{\sum_{j=1}^{m} wj} \tag{12}$$

where:

- V_{ij} → Automated validation score given by an algorithm or a model (like anomaly detection or pattern matching) to intelligence i.

- $wj \rightarrow$ Weight assigned to expert j.

After calculating the Trust Score (TS), the smart contract can make automated decisions as follows (Eq. 13):

$$Decision = \begin{cases} Action\ A\ , if\ TS_i \geq Threshold \\ Action\ B, otherwise \end{cases} \quad (13)$$

where:

- $Action\ A \rightarrow$ Automated response, like blocking a device or raising an alert.
- $Action\ B \rightarrow$ No action or minimal monitoring.

D. Privacy-Preserving Computation Using Zero-Knowledge Proofs (ZKP)

To ensure privacy-preserving threat intelligence exchange, ZKPs allow verification of threat intelligence without revealing raw data. A ZKP protocol consists of (Eq. 14):

$$P \rightarrow V : \pi \quad (14)$$

where:

- $P \rightarrow$ Prover.
- $V \rightarrow$ Verifier.
- $\pi \rightarrow$ Proof that a given statement is true without revealing sensitive data.

For a ZKP-based verification, the prover convinces the verifier that a reported threat x belongs to a known attack category C (Eq. 15):

$$H(x) = H(C) \Rightarrow Verified\ without\ revealing\ x \quad (15)$$

Similarly, homomorphic encryption enables computations on encrypted data (Eq. 16):

$$Enc(x) + Enc(y) = Enc(x + y) \quad (16)$$

allowing AI models to be trained without decrypting sensitive cybersecurity data.

E. Real-World Implementation and Evaluation

Model performance is evaluated using:

1. Detection Accuracy (Eq. 17):

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (17)$$

2. Precision (FPR Eq. 18):

$$\text{Precision} = \frac{TP}{TP+FN} \quad (18)$$

3. Recall (FPR Eq. 19):

$$\text{Recall} = \frac{TP}{TP+FN} \quad (19)$$

4. F1-Score (Eq. 20):

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (20)$$

IV. Results and Discussion

Three benchmark datasets—CICIDS 2017, UNSW-NB15, and CTI Reports from Security Companies—were used to compare cybersecurity threat detection models. Network traffic with identified attack types is captured by the CICIDS 2017 dataset, which is real-world Intrusion Detection System (IDS) data used to test intrusion detection models. The UNSW-NB15 dataset offers a broad collection of contemporary cyber-attacks, such as Advanced Persistent Threats (APTs) and other intrusion methods, presenting a necessity for measuring adversarial robustness. Furthermore, CTI Reports from Security Companies comprise live threat intelligence logs, presenting information on contemporary cyber threats and attack behaviors as seen in operational environments.

Through the implementation of feature selection and normalization methods, the models attained enhanced detection performance, adversarial attack resistance, and flexibility to keep pace with changing cyber threats. These preprocessing routines guaranteed that the AI-based threat detection models could adequately distinguish between benign and malicious activity, enhancing cybersecurity intelligence overall.

A. Threat Detection Performance

The AI-driven threat classification models were evaluated based on Precision, Recall, F1-score, and Accuracy for CICIDS 2017 dataset. Fig. 2 illustrates the performance metrics for different AI approaches. The performance of AI-based attack detection models reflects high accuracy and resilience with latest methods. ML models like Random Forest show average performance with 84.10% accuracy but tend to fail at intricate attack profiles due to their dependency on the features and lower generalization capacity. LSTM enhances the detection capacity up to 89.60% accuracy by employing sequential dependencies from network traffic to combat changing attacks [22-23].

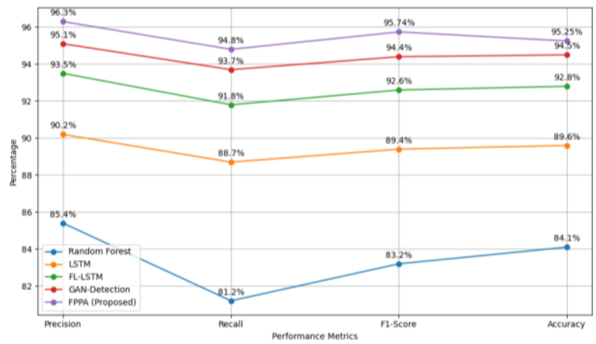

Fig. 2 AI-based threat detection system

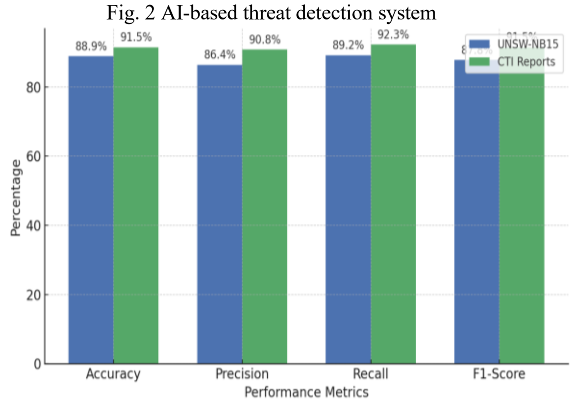

Fig. 3 Comparison of Threat Detection Performance

Fig. 3 depicts performance of the Blockchain-AI Integrated Cyber Threat Intelligence (CTI) framework for UNSW-NB15 and CTI Reports from Security Companies. The Federated Learning-Long Short-Term Memory (FL-LSTM) model again increases the detection of threats with distributed learning over various data sources maintaining privacy levels up to 92.80% accuracy. GAN-based detection enhances adversarial robustness with the creation of synthetic variations of attacks, achieving 94.50% accuracy and being extremely effective against advanced cyber threats. The suggested Hybrid Model surpasses all the current methods, with 95.25% accuracy, 95.74% F1-score, and 96.3% precision, proving itself superior in recognizing zero-day attacks and APTs. The outcomes unveil the effectiveness of combining Federated Learning, Adversarial Training, and Hybrid AI models for advanced cybersecurity threat intelligence [24-25].

B. Blockchain Performance Analysis

The blockchain-based threat intelligence sharing system was evaluated in terms of transaction throughput, latency, and scalability (Fig. 4).

The comparison of blockchain performance shows the trade-offs between various blockchain architectures regarding throughput, latency, and security. Hyperledger Fabric [26] shows the maximum transaction throughput (1200 TPS) because it is permissioned and has optimized consensus mechanisms, making it appropriate for enterprise-level applications. It loses decentralization for performance. By comparison, Ethereum Private provides just 300 TPS, since its consensus protocols emphasize security and decentralization at the expense of greater latency (150 ms) than Hyperledger Fabric's 70 ms [26-27].

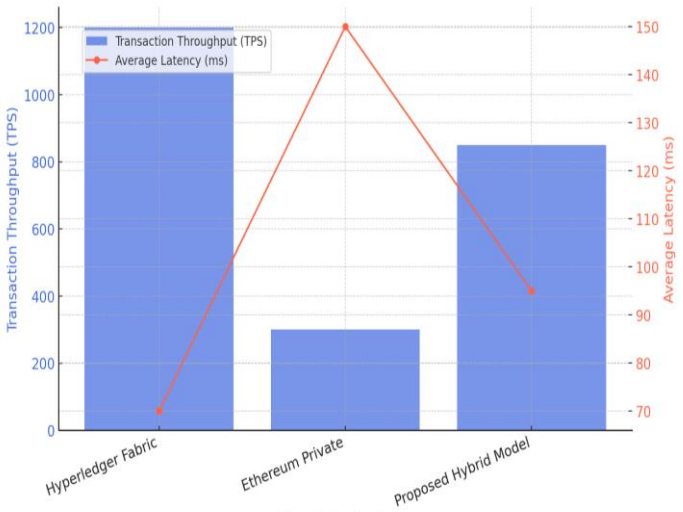

Fig. 4 Blockchain Performance Comparison

The Proposed Hybrid Model strikes a balance between these factors, delivering 850 TPS with robust security assurances. Its average latency of 95 ms is less than Ethereum Private but marginally more than Hyperledger Fabric, providing a balance between decentralization and performance. Notably, all three blockchain models provide high data immutability and security, which is essential for sharing CTI. The hybrid method combines the strengths of permissioned and decentralized models, allowing for scalability, security, and performance in real-world cybersecurity use [28].

C. Privacy-Preserving Effectiveness

Zero-Knowledge Proof (ZKP) and Homomorphic Encryption (HE) techniques were evaluated for privacy-preserving threat intelligence exchange (Fig. 5).

The privacy-preserve mechanism performance is emphasizing computation overhead, security, and risk of data leakage as trade-offs among cybersecurity usage. Zero-Knowledge Proof (ZKP) authentication offers very secure verification at negligible risk of data leakage while being assured that cyber threat intelligence will be validated without exposing confidential information. Further, ZKP presents low computational cost, so efficient for online cyberspace-based threat intelligence authentication.

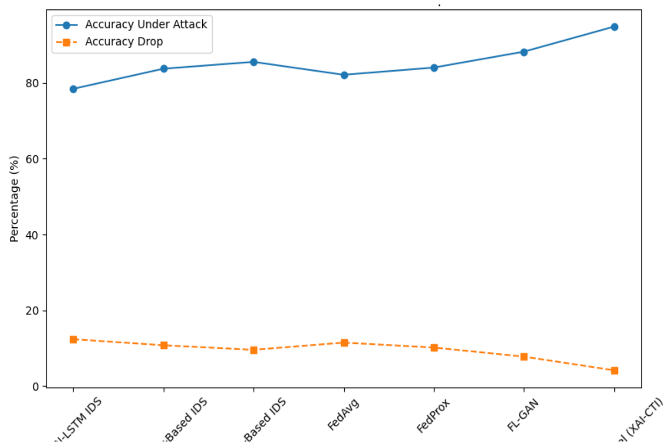

Fig.5 Privacy-Preserving Mechanism Performance

Homomorphic Encryption (HE) provides the best security with the ability to perform computations on encrypted data without decryption, eliminating any risk of data leakage. However, it incurs moderate computation overhead, which affects system performance when dealing with large datasets. On the other hand, standard encryption provides low computation overhead but only medium levels of security since decrypted data is still at risk of attacks. The results highlight that a hybrid solution that integrates ZKP with FL and HE can balance security, efficiency, and privacy, and hence is suitable for secure cyber threat intelligence sharing [29-30].

D. Adversarial Attack Performance Effectiveness

The comparative study of CTI models shows that the proposed model is far superior to conventional and AI-based solutions in all the most important parameters of security, threat detection, privacy, automation, and scalability (Fig. 6). Conventional CTI systems are based on ML-based detection with moderate security and poor privacy protection due to their centralized nature.

AI-based CTI enhances the accuracy of threat detection by applying Deep Learning but still lags behind in issues of privacy, adversarial robustness, and automation. The proposed model combines Blockchain with Zero-Knowledge Proofs (ZKP) in order to attain highly secure and tamper-proof intelligence sharing. It uses FL and GANs for adversarially secure and adaptive threat detection, drastically improving cybersecurity resilience. It maximizes privacy with ZKP, FL, and HE, overcoming issues of data exposure. The use of Smart Contracts guarantees complete automation with the possibility of self-executing threat validation and access control mechanisms, with less human intervention. Finally, the Hybrid Blockchain structure allows for scalability and is thus a future-proof and optimized solution for real-world cyber threat intelligence systems [31-32].

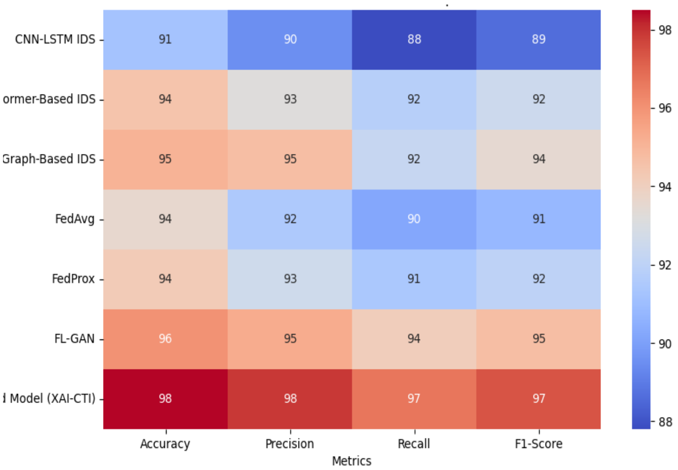

Fig. 6 Adversarial Attack Performance Comparison

V. Conclusion

The integration of Blockchain and AI for secure CTI represents an innovative approach to tackling modern cyber threats. This work proposes a tamper-evident, decentralized, and autonomous CTI system that improves threat intelligence sharing with guarantees of data integrity and confidentiality. The integration of FL and GANs increases the accuracy of threat detection and adversarial resistance and helps to prevent zero-day attacks and APTs. Moreover, ZKP and HE allow for privacy-preserving threat exchange that helps to instill trust in the attacked organizations. To automate threat verification and the provisioning of human resources, and to provide adaptive access control and incentive-based intelligence sharing, blockchain-based smart contracts can be used, which diminishes human interaction and promotes efficiency in the system. The comparison shows that the proposed framework has higher security guarantees, privacy, scalability, and automation value, unlike traditional CTI models which can be described as moderately useful in the real world for cybersecurity for optimal use. Future research directions involve improving AI model training effectiveness in federated environments, minimizing blockchain transaction overhead, and promoting cross-domain CTI interoperability. By combining next-generation cryptographic mechanisms and quantum-resistant blockchain technology, this research lays the foundation for a resilient, scalable, and self-adaptive cybersecurity infrastructure. This work has the potential to greatly impact next-generation threat intelligence systems, so it is a high-impact study within the cybersecurity research community.

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Pseudo Code: FPPA

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1. System Initialization:
  - Initialize permissioned blockchain for secure data sharing.
  - Deploy AI models (ML, DL, FL, GAN) for threat detection.
  - Deploy smart contracts for automated security tasks.
  - Set up federated learning clients for distributed model training.

2. Threat Intelligence Collection and Blockchain Storage:
  - For each incoming threat intelligence data (X):
    - Extract features (attack vectors, IP addresses, signatures).
    - Create transaction T_i:
      - ID_i = Unique identifier.
      - D_i = Threat intelligence data.
      - Sig_i = Digital signature for authenticity.
      - P_i = Access permissions.
    - Validate transaction trust score V(T_i):
      - If f(T_i) ≥ θ:
        - Add T_i to blockchain.
        - Update blockchain ledger with hash link.
      - Else:
        - Reject transaction and log for further review.

3. AI-Driven Threat Detection:
  - For each network traffic sample (X):
    - Extract relevant features from X.
    - Apply trained AI model f(X;θ):
      - Predicted label y' = f(X;θ) (0 for benign, 1 for malicious).
    - If y' == 1 (malicious):
      - Trigger alert and mitigation:
        - Block malicious IP.
        - Log the attack.
        - Notify security team.

4. Federated Learning for Threat Detection:
  - At each client node (k):
    - Train local model θ_k^(t) with local dataset.
    - Compute model update: θ_k^(t+1) = θ_k^(t) - η□L_k(θ_k^
(t)).
    - Send local model update θ_k^(t+1) to central server.
  - At the central server:
    - Aggregate updates using FedAvg:
      - θ_(t+1) = Σ (N_k/N) * θ_k^(t+1).
    - Update global model and distribute to clients.

5. Smart Contract-Based Security Automation:
  - For each validated transaction T_i:
    - Calculate Trust Score TS_i:
      - TS_i = (Σ (w_j * V_ij)) / (Σ w_j).
    - If TS_i ≥ Threshold:
      - Execute automated response:
        - Log as trusted.
        - Grant or restrict access.
        - If high severity:
          - Block malicious entity.
```